UNSW MARKETING ANALYTICS HACKATHON 2026

Beyond a good story

What 1.45 million Kiva loan descriptions
can and cannot tell us about funding speed







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Text carries forecast information, but no writing tip is proven

Summary across the explanatory models, robustness checks and forecast benchmark

EXPLANATORY MODELS

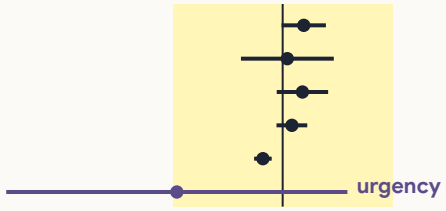
64 of 120

Uncertainty first

Yellow squares: significant terms that lost significance once errors were clustered by 48 countries.

EQUIVALENCE TEST, EXPLORATORY

±5%

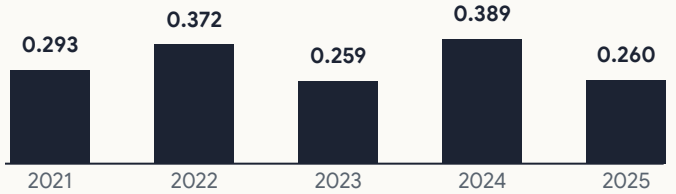


Most framing links are small

Five framings stay inside the ±5% band per extra mention per 100 words; urgency does not.

FORECAST BENCHMARK

5 of 5 years



Text adds forecast information

Text lowered linear forecast error every year, by 0.26–0.39 days; the median is about 7 hours.

Next decision: validate the text measures, then test one truthful prompt with partners.

Validate measures | Close data gaps | Pilot one prompt

Agenda

01

Problem

What are we measuring, and why does it matter?

Editable fields 4

The outcome 5

A slower market 6

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Approach

How did we test the claims?

Four analyses 7

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What holds up, and how well?

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The description is one of the few listing fields still editable

Illustrative listing: what is fixed and what is editable when a loan is listed



FIXED AT LISTING 3

Amount, median	Term, median	Borrowers, median
US$325	**12 months**	**1**
Sector, most common	Country, most common	Partner
Agriculture	**Philippines**	**Field partner**

EDITABLE DESCRIPTION 1

She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow. 4

Highlights mark framing terms counted per 100 words 2

Family | Business | Urgency | Gratitude

Figure 1. Illustrative composite listing, not a real borrower; values are dataset medians and most common categories.

1 The description is the editable part

Most fields are fixed. The **description can still be edited**, so if framing mattered, wording guidance would be a cheap lever.

2 Eight framing rates, every loan

Eight framing rates per 100 words, plus sentiment and text form, across 1,453,840 loans.

3 The test is for whom and when

Which framings are linked to **faster funding**, for which loans, and when?

4 Pronouns reflect who writes

Descriptions are mostly **written about the borrower, not by them**, so pronouns may reflect partner conventions.

The outcome is platform funding time, not borrower waiting time

Descriptive audit of all records



Loans disbursed to the borrower before listing



Who is in the data

In the data: 1,453,840 completed loans, including 1,637 refunded

Not in the data: listings that expired or never funded 2

Figure 2. Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; this step is not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

Scope: 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

1 Partners wait, not borrowers

96.4% of loans reached the borrower a median 24.2 days before listing; platform time plausibly reflects replenishing partner capital.

2 Unfunded loans are missing

Only loans that **eventually funded** are in the data, so we cannot study whether a loan funds at all.

3 We measure platform speed

Our outcome runs from **listing to full funding**, median 2.85 days; it is not funding success or borrower welfare.

Same-day funding fell from 46.0% to 30.3% and stayed low through 2024

Share of loans fully funded within 24 hours, by posting year; descriptive

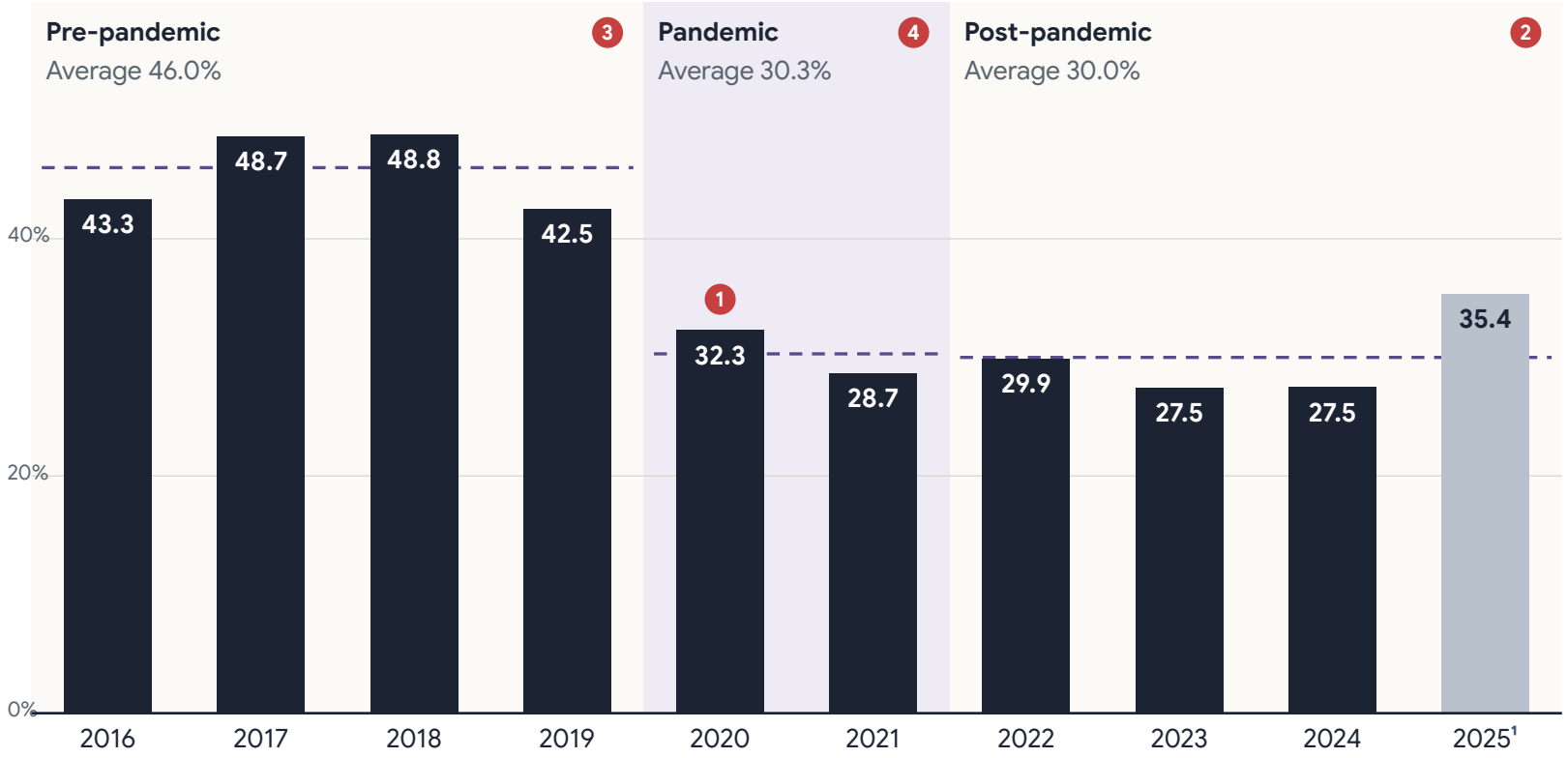


Figure 3. Share of completed loans funded within 24 hours, by posting year, with period averages.

[1] 2025 may be incomplete: loans still unfunded at extraction are missing.

1 The drop came in one year

Same-day funding fell from **42.5% in 2019 to 32.3% in 2020**.

2 The low level held four years

From 2021 to 2024 the share stayed **between 27.5% and 29.9%**; the 2025 rise may reflect an incomplete year.

3 Every model controls for period

The shift is platform-wide, so **narrative effects are compared within periods**, not across them.

4 Causes are untested here

Possible drivers, such as lender supply or listing volume, are **not measured** in this analysis.

Four analyses, each answering a different question

One dataset; the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries

Explanatory models → 8

Which features are linked to funding speed, and how sure are we?

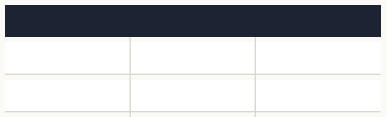
Regressions of funding time and same-day funding; errors clustered by country.

64 of 120 significant terms lost significance

Robustness checks → 9

Do the links survive richer controls?

The same models with more text measures and a programme proxy; multiple-test adjusted.

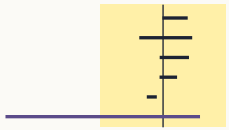


Only small associations remain

Equivalence test → 10

Which links are small enough to rule out a large effect?

Is each 90% interval inside ±5% of funding time?



Five framings small; urgency unresolved

Forecast benchmark → 11

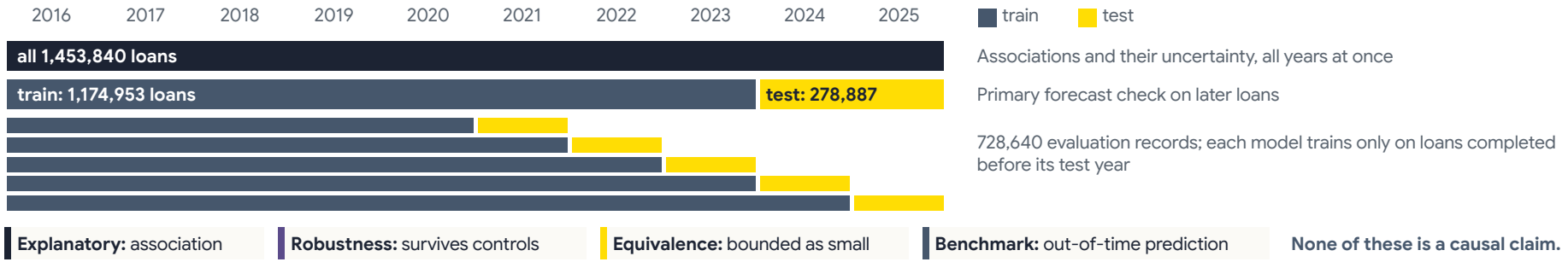
Does text improve a forecast built on loan structure?

Train on earlier years, then test each year with and without text.



Error fell in 5 of 5 years, by 0.26–0.39 days

Years each analysis uses



What each can claim

- **Explanatory:** association
- **Robustness:** survives controls
- **Equivalence:** bounded as small
- **Benchmark:** out-of-time prediction

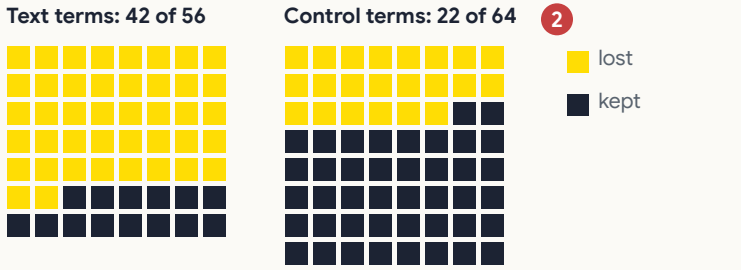
None of these is a causal claim.

Large row counts did not remove country-level uncertainty

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

(a) Which terms lost significance

64 of 120 terms significant under HC3 lost it under clustering; each square is one term.



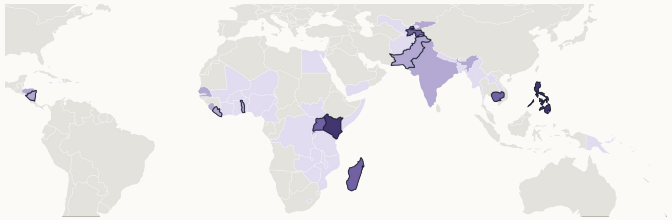
(b) Urgency before and after

HC3 errors allow unequal variance but treat every loan as independent; clustering by country lets loans in a country share shocks.



(c) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.



	Country	Region	Loans	Share	Same-day
1	**Philippines**	Asia	521,586	35.9%	48%
2	**Kenya**	Africa	266,562	18.3%	29%
3	**Uganda**	Africa	69,651	4.8%	31%
4	**Tajikistan**	Asia	68,865	4.7%	18%
5	**Cambodia**	Asia	58,074	4.0%	49%
6	**Madagascar**	Africa	55,670	3.8%	67%
7	**Pakistan**	Asia	42,767	2.9%	15%
8	**Nicaragua**	Central America	38,520	2.6%	15%
9	**Liberia**	Africa	28,847	2.0%	28%
10	**Togo**	Africa	27,513	1.9%	38%

Figure 4. Explanatory models, HC3 versus errors clustered by country (48 clusters); map and table: share of all loans and of loans funded within 24 hours, by country.

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

1 Text terms are hit hardest

42 of 56 text terms lost significance, against 22 of 64 controls; 42 of the 56 terms that kept it are controls.

2 Few text terms survive

Only **sentiment and some family-framing interactions** with sector or region stay significant.

3 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

4 Few countries dominate

Ten countries hold **81% of loans**, Asia and Africa 92.8%; regional patterns rest on **two countries each**, so no regional writing rules yet.

Few narrative associations remain after programme controls

Robustness checks: country-clustered, BH-adjusted within specification

Term and outcome	Expanded	Programme-adjusted
Third person, log(1 + days)	−0.0090, BH p 0.012	−0.0081, BH p 0.061
Business, 24-hour (pp)	not significant after BH	−0.258 pp, BH p 0.006

Figure 5. Per extra mention per 100 words; BH-adjusted within each specification.

0.012 → 0.061 3 1

third-person BH p, expanded to programme-adjusted

−0.258 pp 2

business framing, same-day rate, programme-adjusted

1 Only small associations remain

Third-person framing keeps its size but rises to **BH p 0.061**; business framing is **the one term below BH 0.05**.

2 These are still associations

Programme text is an **unvalidated proxy**; neither result shows what editing would do.

3 Pronouns may reflect the writer

Descriptions are usually written by partners, so the **third-person link may reflect who writes**, not a borrower's choice.

Five framings have small links to funding time; urgency remains unknown

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

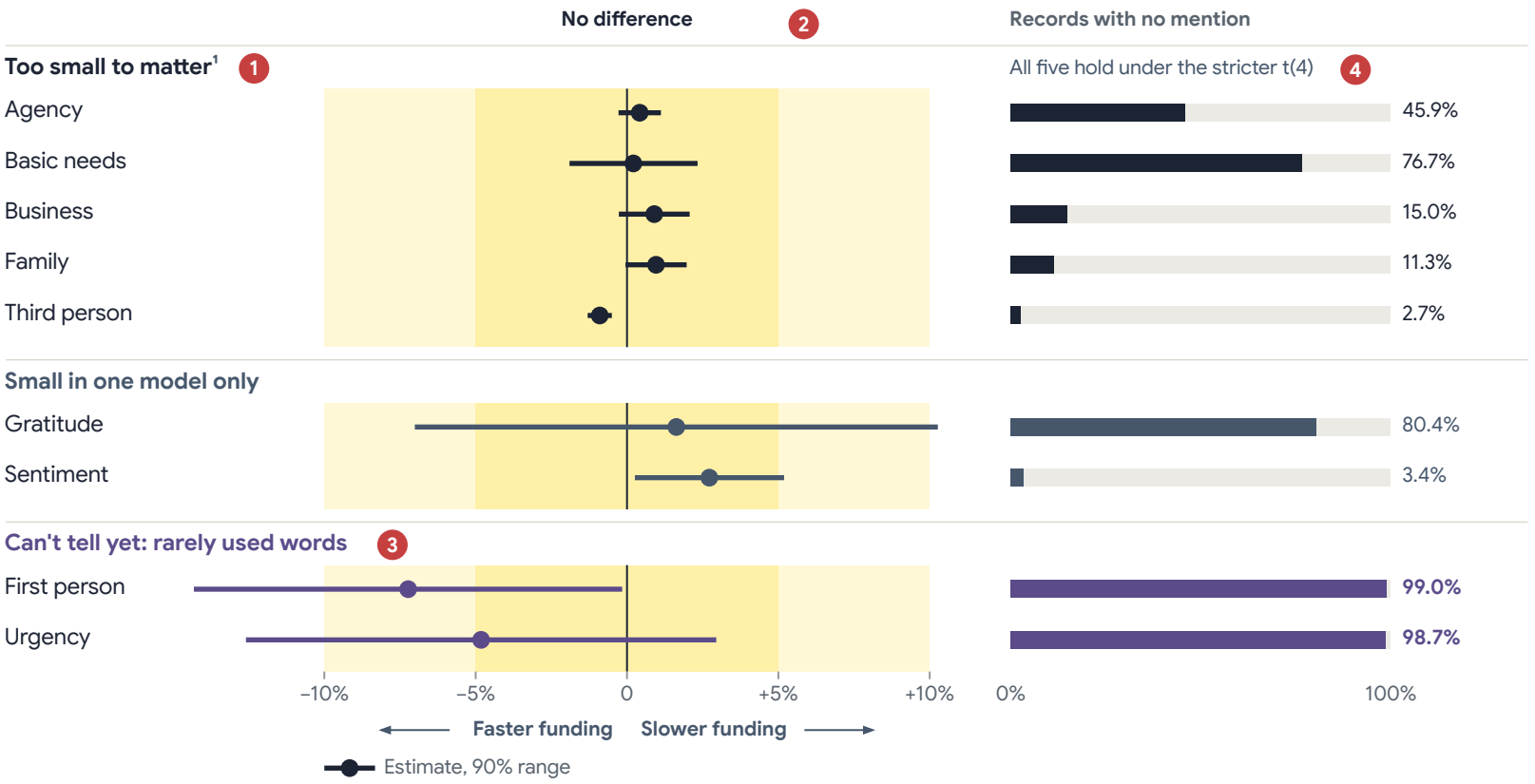


Figure 6. Change in funding time per extra mention per 100 words, expanded model, 1,453,840 loans.

[1] Within ±5% of funding time: ±0.05 on log(1 + days), BH-adjusted; bound set after the estimates.

1 Five framings have small links

Family, business, basic needs, agency and third person stay **within ±5%** in both models.

2 Small means under five hours

±5% is about **4.5–4.7 hours** on a typical 2.85-day loan.

3 Urgency is too rare to judge

Under **2% of descriptions** use urgency or first person; their ranges still allow **6–14%** either way.

4 The verdicts hold under stricter tests

All five pass a **stricter t(4) test**; three stay small even within **±2%**.

Adding text lowered forecast error in 5 of 5 years

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

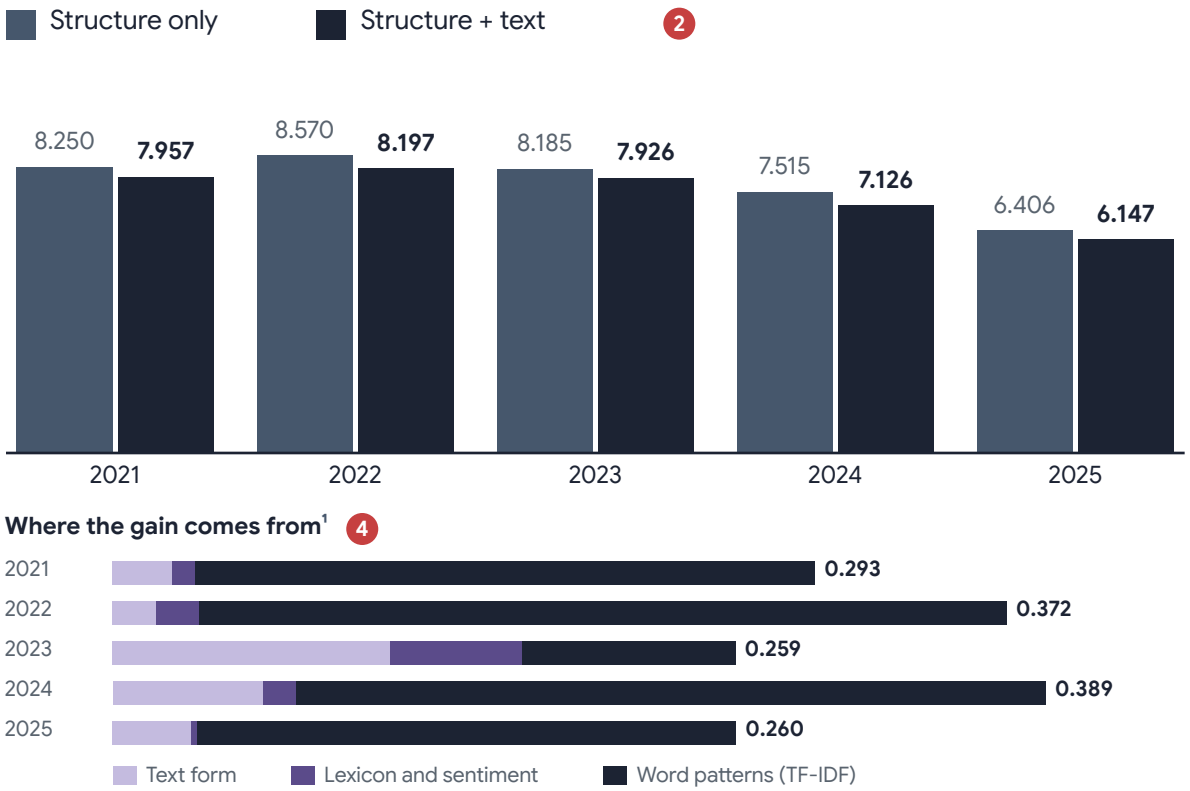


Figure 7. Linear-model MAE in days (top) and its reduction by text feature group (bottom), identical evaluation rows.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

0.29 days 1

median annual error reduction

5 of 5 3

years with lower error

728,640

evaluation records

1 Text carries forecast information

Gains of **0.26–0.39 days** on 728,640 evaluation records.

2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

4 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add only **0.003–0.055 days**.

The gain is modest and depends on the model

Forecast benchmark: annual MAE gain with month-resampled 95% intervals

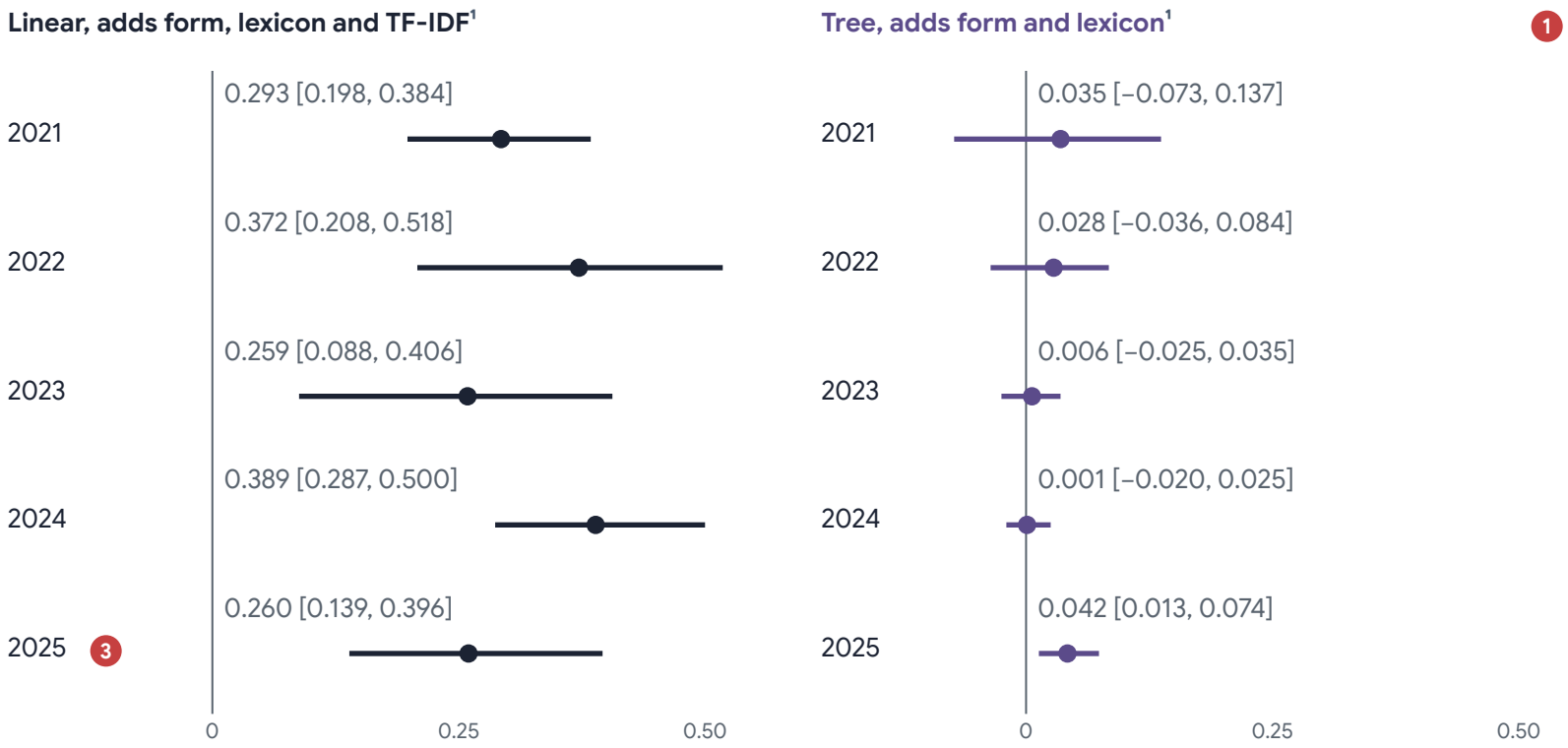


Figure 8. Annual MAE gain in days with month-resampled 95% intervals; positive is better.

2 [1] Country resampling, not shown: intervals include zero in 4 of 5 linear and 3 of 5 tree years.

1 The tree model gains little

The tree model, without TF-IDF, gains **at most 0.04 days**.

2 Country intervals are wide

Country resampling **includes zero in 4 of 5** linear years. Capping extreme forecasts, an exploratory check, leaves zero in 1 of 5.

3 Metrics move in different directions

In 2025 text improved error and calibration but **AUC fell** from 0.893 to 0.888.

What the evidence supports, and what it does not

Synthesis across all layers

Supported	Not established
Text adds forecast information in a linear model	That editing a description speeds funding
Most framing associations are small	Any country or regional writing rule: regional patterns rest on two countries each
Platform funding time is not borrower waiting time for most loans	Effects of urgency or first-person framing
	Performance on loans that never funded

Three gates before any writing advice

Proposed next research for Kiva and participating partners; each gate must pass before the next

01 → 4

Validate the measures

Question

Do our word counts capture clarity, need and agency?

Do

Build a small codebook; two people code the same narratives and resolve disagreements.

Done when

Definitions are frozen before later periods are checked.

02 → 5

Close the data gaps

Question

What did lenders see, and which listings never funded?

Do

Get original texts, partner identifiers, exposure data and expired listings.

Done when

Unfunded listings and listing-time text are in the data.

03 → 13

Pilot one prompt

Question

Does editing a description change funding speed?

Do

With partners, test one truthful clarity prompt against a control.

Done when

Threshold, power and guardrails are agreed in advance.

Each gate unlocks a stronger claim

Today
Associations and forecast information

→

After gate 1
Measures that mean what we say

→

After gate 2
Results for every listing, not only funded ones

→

After gate 3
A tested effect of editing

If a gate fails, stop there: report the uncertainty and give no writing advice.

Thank you

Mike Nguyen and Sophia Nguyen, Cultural Blend

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



Regional family links look strong but rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals



Countries behind each region 3

Africa	27	Central America	2
Asia	12	Middle East	2
Oceania	4	North America	1

Figure A1. Within-region family-framing slopes, 95% ranges, explanatory models; one dot per country, Kiva regions.
[1] t(1) reference on the same clustered error: a heuristic screen, not a calibrated test. Axis in log points × 100.

1 Both links look strong at first

Standard errors put family framing at about **7% faster funding** per extra mention in each region.

2 Two countries cannot carry them

A few-country check widens both ranges **across zero**; the Middle East is **97% Palestine**.

3 Other regions have many more countries

Africa has **27 countries** and Asia **12**; these two regions have two each.

4 No regional writing rules yet

Treat both as **descriptive patterns**; a real test needs more countries or partner-level data.

As-of training audit

Forecast benchmark: training records must have completed before each cutoff

Evaluation year	Pre-cutoff postings completing later	Eligible as-of training	Evaluation rows
2021	7,528	717,672	163,166
2022	3,918	884,448	143,354
2023	3,546	1,028,174	143,233
2024	5,691[1]	1,169,262	141,725
2025	4,261	1,312,417	137,162

Figure A2. Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

Chronological holdout metrics

Explanatory models: chronological holdout, completed loans only

5.2031 days

duration MAE, HistGradientBoosting

0.5420

duration R^2

0.8997

24-hour ROC AUC

0.8301

24-hour average precision

0.1156

24-hour Brier score

Scope: 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

Exploratory story themes

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

1.5 days

mean funding time, fastest theme (sanitation)

13.5 days

mean funding time, slowest theme (group solar and farm plots)

What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.